

Project Proposal: Generative Depth Completion for Transparent Objects

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1. Problem Definition

Accurate depth perception is a cornerstone of autonomous robotic manipulation, enabling tasks such as grasping, path planning, and collision avoidance. However, standard RGB-D sensors (e.g., RealSense, Kinect) fail catastrophically when encountering transparent or highly reflective objects—such as plastic bottles, glassware, or polished metal parts common in industrial manufacturing cells. Due to the physics of light refraction and specular reflection, these sensors produce depth maps riddled with "holes," noise, and varying artifacts, rendering the data unusable for precise interaction.

The scientific problem is that the missing depth information cannot be recovered by simple deterministic interpolation or filtering, as the geometry is lost. This project frames the problem as a **conditional generation task**: given the RGB context (which clearly shows the object) and the incomplete depth map, can we generate a statistically plausible and geometrically accurate completion of the depth map?

2. Solution and Hypothesis

I propose a **Pixel-Space Conditional Denoising Diffusion Probabilistic Model (cDDPM)** to perform depth inpainting. Unlike discriminative regression models (e.g., standard U-Nets trained with MSE loss) which tend to produce blurry outputs in uncertain regions, a diffusion model learns the explicit data distribution of valid depth geometries.

Hypothesis: A generative diffusion model conditioned on both the RGB image and the raw (noisy) depth sensor data will outperform standard interpolation techniques and regression-based baselines in recovering the geometry of transparent objects. By learning the joint distribution of visual appearance (RGB) and geometry (Depth), the model can "hallucinate" the correct curvature and fill in the sensor holes consistent with the object's visual features.

3. Experiment Plan

The project will be executed in three phases, optimized for our term timeline and limited computational resources (RTX 4060 Mobile, 8GB VRAM):

1. Data Preparation:

- I will curate a lightweight training subset from the **ClearGrasp** dataset (approx. 5k-10k images).
- Images will be resized to 128×128 or 256×256 to ensure the model fits within

GPU memory during training.

- Depth values will be normalized to the range $[-1, 1]$ to match the standard diffusion process.

2. Model Implementation:

- I will base my code on the **lucidrains/denoising-diffusion-pytorch** repository, extending the standard U-Net architecture.
- **Architecture:** A Conditional U-Net operating in pixel space. The input will be modified to accept a 4-channel conditioning tensor (3 channels for RGB + 1 channel for the raw/incomplete depth) concatenated with the noisy target.
- The network will predict the noise residual added to the "clean" ground truth depth.

3. Evaluation & Future Work:

- **Quantitative:** I will calculate Root Mean Square Error (RMSE) and Structural Similarity Index (SSIM) on the test set, specifically masking the evaluation to the transparent object regions.
- **Qualitative:** I will visually compare the generated depth maps against the raw sensor input and a zero-shot baseline using the pre-trained **Marigold** model (running in inference-only mode).
- **Future Work:** While outside the scope of this term project, the ultimate goal is to integrate this trained model into my Master's thesis digital twin to enable robust grasping of transparent parts in a jigless manufacturing cell.

4. Dataset

I will use the **ClearGrasp** dataset. It is uniquely suited for this research because it provides a perfect paired ground truth for real-world transparent objects. The authors created this dataset by capturing a scene with transparent objects and then physically swapping them with identical spray-painted (opaque) counterparts to capture the "true" depth.

- **Input:** RGB Image + Raw Depth (from RealSense D435, containing holes).
- **Target:** Ground Truth Depth (from the opaque swap).

This dataset allows for fully supervised training of the generative model while addressing a real-world domain gap in robotics.